

Alkaline diets favor lean tissue mass in older adults

Background Maintaining muscle mass while aging is important to prevent falls and fractures. Metabolic acidosis promotes muscle wasting, and the net acid load from diets that are rich in net acid-producing protein and cereal grains relative to their content of net alkali-producing fruit and vegetables may therefore contribute to a reduction in lean tissue mass in older adults. **Objective** We aimed to determine whether there was an association of 24-h urinary potassium and an index of fruit and vegetable content of the diet with the percentage lean body mass (%LBM) or change in %LBM in older subjects. **Design** Subjects were 384 men and women ≥ 65 y old who participated in a 3-y trial comparing calcium and vitamin D with placebo. Potassium was measured in 24-h urine collections at baseline. The %LBM, defined as $\text{total body nonfat, nonbone tissue weight} \div \text{weight} \times 100$, was measured by using dual-energy X-ray absorptiometry at baseline and at 3 y. Physical activity, height, and weight were assessed at baseline and at 3 y. **Results** At baseline, the mean urinary potassium excretion was 67.0 ± 21.1 mmol/d. Urinary potassium (mmol/d) was significantly positively associated with %LBM at baseline ($\beta = 0.033$, $P = 0.006$; adjusted for sex, weight, and nitrogen excretion) but not with 3-y change in %LBM. Over the 3-y study, %LBM increased by $2.6 \pm 3.6\%$. **Conclusion** Higher intake of foods rich in potassium, such as fruit and vegetables, may favor the preservation of muscle mass in older men and women.